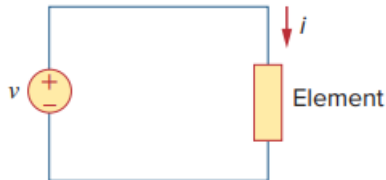


Problem

In Fig. 6.43, if $i = \cos 4t$ and $v = \sin 4t$, the element is:

- (a) a resistor
- (b) a capacitor
- (c) an inductor

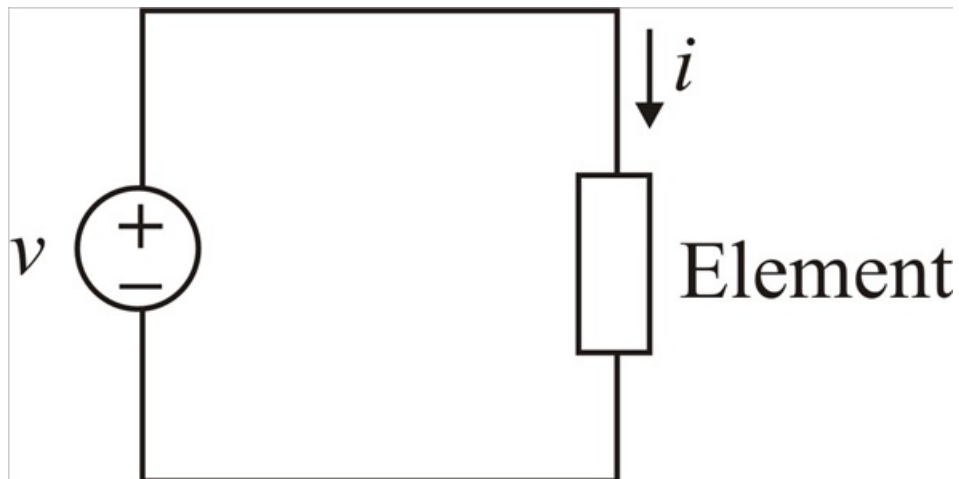
Fig. 6.43:



Step-by-step solution

Step 1 of 2

Given circuit diagram is



Step 2 of 2

Given that

$$i = \cos 4t \text{ and}$$

$$v = \sin 4t.$$

Using Ohm's law to the above circuit

$$\begin{aligned} \text{Element} &= \frac{v}{i} \\ &= \frac{\sin 4t}{\cos 4t} \\ &= \frac{\cos(4t - 90^\circ)}{\cos 4t} \quad \left(\text{Since, } \mp \sin x = \cos(x \pm 90^\circ) \right) \\ &= \frac{\cos(4t - 90^\circ)}{\cos 4t} \end{aligned}$$

Here, we have phase shift of -90° . So given element is a **capacitor**.